Youcef Melakhessou

Seeking a 4 to 6-month internship starting March 29, 2027, in the field of Data, as a Data Analyst, Data Scientist, or on Business Intelligence (BI) oriented assignments

Contact details

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Languages

- Arabic : **Native language**
- English : **Intermediate (B2)**
- French : **Fluent**

Soft skills

- Organizational skills
- Teamwork
- Analytical thinking

Tools

- Visual Studio
- PowerBI
- Talend
- GitHub
- Anaconda

Interests

- Automotive
- Travel
- Gaming
- Swimming

Profile

Currently a Master's student in Decision Information Systems at the University of Lorraine, I am particularly interested in data analysis and data valorization. I am currently looking for an internship in the field of Data, in order to apply my skills and contribute to data-oriented and decision-support projects.

Formation

Master's Degree in Decision Information Systems *2025-2027*
University of Lorraine - UFR MIM Metz

Bachelor's Degree in Computer Science *2022-2025*
University of Lorraine - UFR MIM Metz

Research intership

LCOMS Laboratory University of Lorraine *April-June 2025*

- Comparative analysis of community detection methods on graphs based on scientific papers and real and synthetic datasets.
- Construction of a taxonomy matrix to compare the performance, accuracy, scalability, and complexity of the approaches studied.
- Design of new local propagation strategies based on the selection of central nodes, multi-label propagation, and community core extraction.
- Study of Deep Learning approaches for graphs (GCN) and their integration with propagation methods.
- Analysis and visualization of results using Python / Jupyter Notebook, using metrics such as NMI and ARI and a computing environment with Spark/GPU.

Programming Languages

- **Python** : Good level
- **SQL** : Proficient
- **DAX** : Proficient
- **C** : Good level
- **C++** : Good level
- **Java** : Proficient
- **JavaScript** : Good level
- **TypeScript** : Good level

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Academic projects

Oracle Database Design and Management Project

February-April 2025

- Design of a relational database under Oracle for the management of projects, employees, equipment, and costs.
- Implementation of business rules and integrity constraints in PL/SQL to ensure data consistency.
- Development of PL/SQL triggers and procedures to automate controls and processing during INSERT, UPDATE, and DELETE operations.
- Automation of employee costs, equipment costs, and monthly cost calculations, as well as the monitoring of usage hours.
- Implementation of budget alerts in case of a project budget overrun.

Research Initiation Project

December-May 2026

- Design and implementation in Java of three approaches for solving the Le Compte est Bon game: DFS with pruning, simulated annealing, and a hybrid approach.
- Design of a Chain of Responsibility architecture to modularize pruning techniques and analyze their impact on the search space.
- Experimental study and benchmark of the different approaches: execution time, number of explored nodes, and success rate.
- Development of a hybrid approach combining Simulated Annealing and DFS, reducing the search space from 21 trillion to 12,317 operations for n=12, in less than one second.

Artificial Intelligence Project

January-April 2026

- Design of a hybrid game-playing agent combining supervised learning, heuristics, and MinMax search with Alpha-Beta pruning.
- Development and training of an MLP neural network on 303 expert decisions to learn move choices.
- Preparation and normalization of input data, followed by evaluation of the model's influence on decision-making.
- Implementation of a weighted decision function combining the outputs of the MLP, heuristic, and MinMax.
- Experimental testing and tuning of hyperparameters/coefficients to improve the bot's stability and performance.